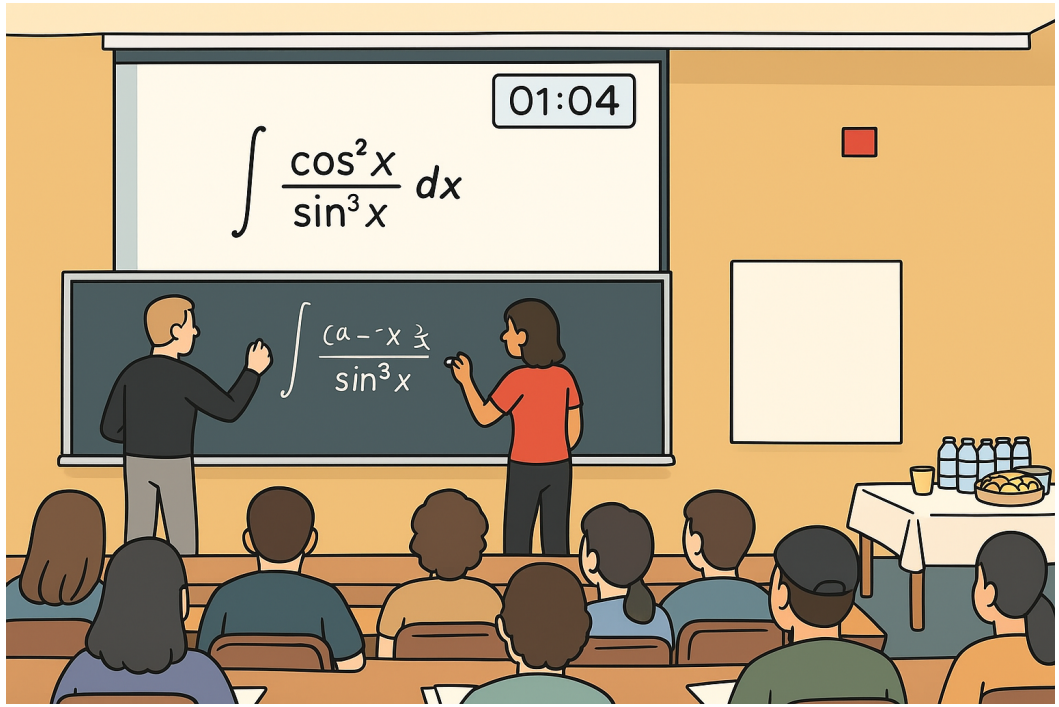
Integratión Bee Physics
Cuartos de Final



PRIMER TORNEO DE INTEGRALES
INTEGRATION BEE PHYSICS
CARRERA DE FISICA
UMSA

Cuartos de Final

GRUPO 1

Integral #3 **Integratión Bee Physic**

$$\int_0^{\frac{1}{3}} \frac{(4-6x)}{1+\frac{3}{2}x+\frac{9}{4}x^2+\frac{27}{8}x^3+\cdots}\,\mathrm{d}x$$

Respuesta Integral #3 **Integratión Bee Physic**

$$\int_0^{\frac{1}{3}} \frac{(4-6x)}{1+\frac{3}{2}x+\frac{9}{4}x^2+\frac{27}{8}x^3+\cdots}\mathrm{d}x = \frac{7}{9}$$

Integral #46 **Integratión Bee Physic**

$$\int \sec^3 x \, \mathrm{d}x$$

Respuesta Integral #46 **Integratión Bee Physic**

$$\int \sec^3 x \, \mathrm{d}x =$$

$$\frac{1}{2}\sec(x)\tan(x) + \frac{1}{2}\ln\left|\sec(x) + \tan(x)\right| + C$$

Integral #21 **Integratión Bee Physic**

$$\int_{-\frac{\pi}{2}}^{0} \cos x(\sin x + \cos x)^{10}\,\mathrm{d}x$$

Ayuda: escriba $\sin x + \cos x$ como una sola función trigonométrica

Respuesta Integral #21

Integratión Bee Physic

$$\int_{-\frac{\pi}{2}}^{0} \cos x(\sin x + \cos x)^{10}\,\mathrm{d}x = \frac{1}{11}$$

GRUPO 2

Integral #68 **Integratión Bee Physic**

$$\int_0^{\pi} \frac{x \sin x}{1+\sin^2 x} \mathrm{d}x$$

$$C.V \rightarrow \quad x = \pi - t$$

Respuesta Integral #68

Integratión Bee Physic

$$\int_0^{\pi} \frac{x \sin x}{1+\sin^2 x}\, \mathrm{d}x =$$

$$\frac{\sqrt{2}}{4}\pi \ln\left(\frac{1+\sqrt{2}}{\sqrt{2}-1}\right)$$

Integral #73 **Integratión Bee Physic**

$$\int \frac{\sin x + \cos x}{9 + 16 \sin 2x} \mathrm{d}x$$

Respuesta Integral #73 **Integratión Bee Physic**

$$\int \frac{\sin x + \cos x}{9 + 16 \sin 2x} \, \mathrm{d}x =$$

$$\frac{1}{40} \ln \left| \frac{5 - 4(\sin x - \cos x}{5 - 4(\sin x - \cos x} \right| + C$$

Integral #75 **Integratión Bee Physic**

Sí $$\int \frac{\sqrt{4+x^2}}{x^6}\mathrm{d}x =$$

$$\frac{A\left(4+x^2\right)^{\frac{3}{2}}\left(Bx^2-6\right)}{x^5}+C$$

¿Cuánto vale A y B?

$$\int \frac{\sqrt{4+x^2}}{x^6}\mathrm{d}x =$$

$$\frac{1/120\left(4+x^2\right)^{\frac{3}{2}}\left(x^2-6\right)}{x^5} + C$$

$$A = \frac{1}{120} \quad B = 1$$

Integral #19 **Integratión Bee Physic**

$$I = \int_0^{\frac{\pi}{2}} (\sin x)^{11} \, \mathrm{d}x$$

Ayuda: Calcule

$$I_{2n+1} = \int_0^{\frac{\pi}{2}} \sin^{2n+1} x \, \mathrm{d}x \sim I_{2n-1}$$

Respuesta Integral #19 **Integratión Bee Physic**

$$\int_0^{\pi/2} \sin^{11} x \mathrm{d}x = \frac{10}{11} \cdot \frac{8}{9} \cdot \frac{6}{7} \cdot \frac{4}{5} \cdot \frac{2}{3}$$